# KickOff - A Football Tournament Management App

Session 2023-2027


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2023-CS-85

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CSC-410 Mobile Application Development


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## Abstract

KickOff is a Kotlin-based Android application designed to simplify the management of small-scale football tournaments. The application enables users to create tournaments, manage teams, record match results, and view real-time leaderboards. Instead of using a traditional database system, the application implements a lightweight JSON-based storage mechanism using SharedPreferences. It also incorporates role-based access control, ensuring that only the organizer of a tournament can modify its data, while other users can view information. The system emphasizes data integrity, performance, and clean user experience. Features such as cascading updates, dynamic leaderboard calculation, and efficient image handling make the application robust and practical for use in the real-world.

## 1 Introduction

Managing sports tournaments manually can be inefficient and error-prone. KickOff addresses this problem by providing a centralized digital platform for organizing football tournaments.

The application is designed with simplicity and performance in mind, making it suitable for small- to medium-scale tournament management without requiring internet connectivity or a backend server.

## 2 Objectives

- To design a mobile application for managing football tournaments
- To implement role-based access control
- To simulate database behavior using JSON storage
- To ensure data integrity using cascading operations
- To provide a clean and responsive user interface

## 3 Technology Stack

- **Language:** Kotlin
- **Platform:** Android SDK
- **UI Design:** XML + Material Design Components
- **Storage:** SharedPreferences with JSON

# 4 System Architecture

The application follows a modular architecture divided into multiple layers:

## 4.1 Models Layer

Defines data structures such as User, Tournament, Team, Match, and LeaderboardEntry.

## 4.2 Utils Layer

Handles logic such as:

- JSON serialization
- Session management
- Image optimization
- Data storage operations

## 4.3 Activities Layer

Manages UI screens and navigation.

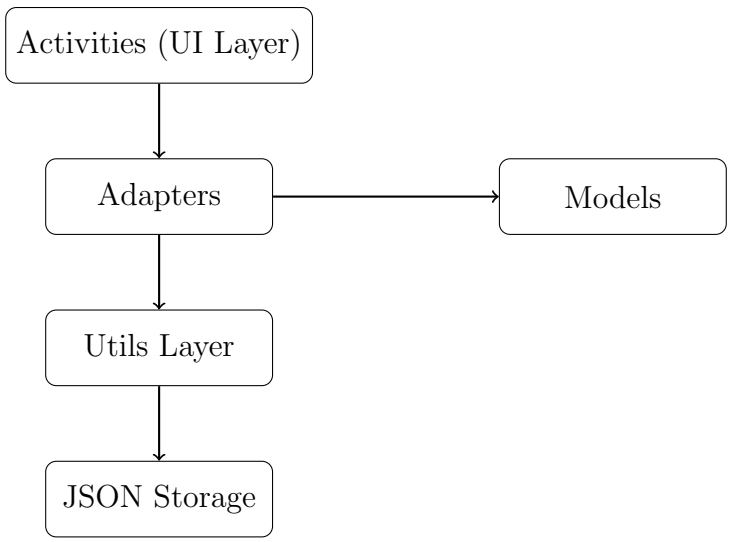
## 4.4 Adapters Layer

Connects data with RecyclerView UI components.

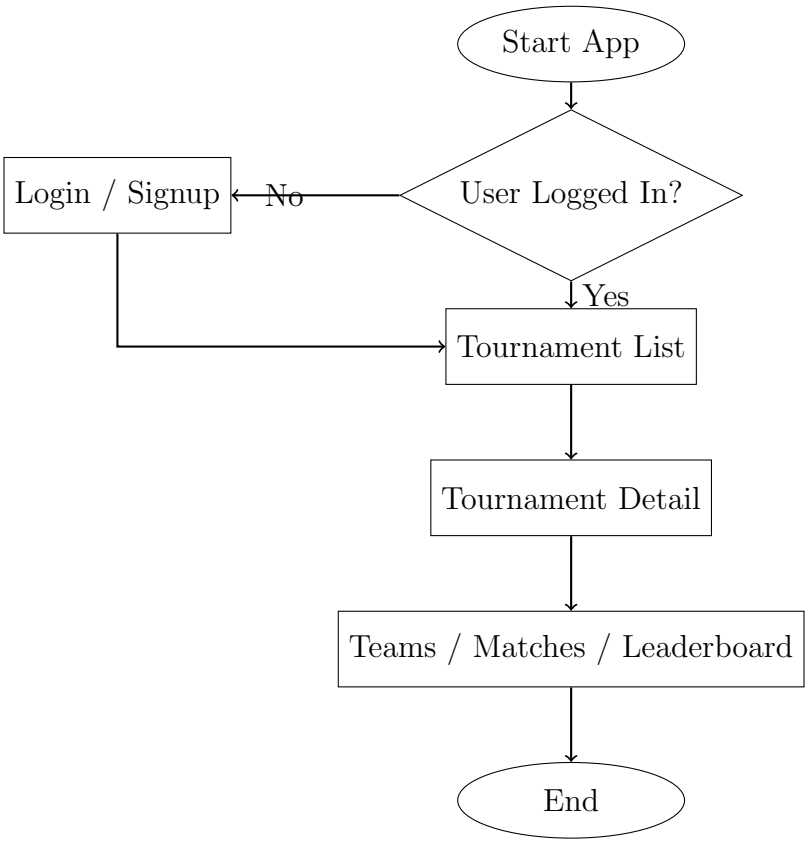
## 4.5 Resources Layer

Contains layouts, colors, strings, and UI assets.

## 4.6 System Architecture Diagram

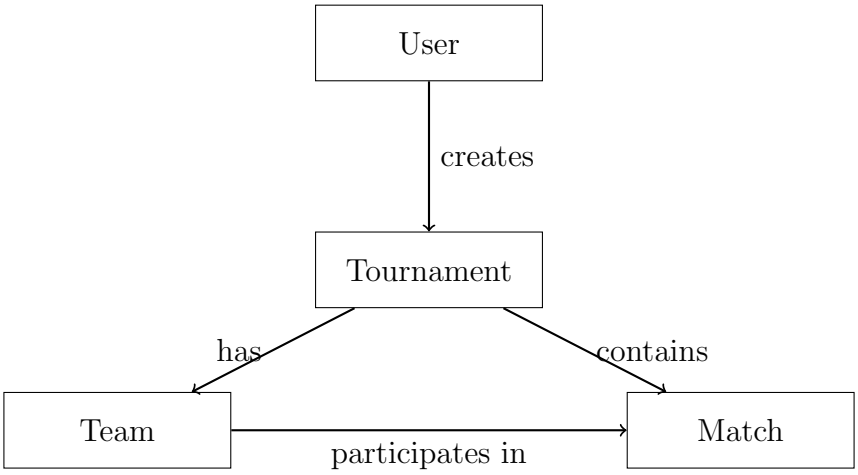


## 5 Application Flowchart




The flowchart illustrates the navigation logic starting from authentication to tournament interaction modules.

## 6 Data Relationship Diagram




The diagram represents logical relationships between entities, simulating a relational database structure using JSON storage.

# 7 Data Storage and Persistence

The application uses a JSON-based storage system instead of a traditional database.

- Data is serialized into JSON format
- Stored in SharedPreferences
- Retrieved and deserialized when needed

This approach keeps the app lightweight and fully offline.

# 8 Data Integrity and Cascading

To maintain consistency, the system implements cascading logic:

- Renaming a team updates all related match records
- Deleting a tournament removes all its teams and matches
- Deleting a team removes associated matches

This prevents orphan data and ensures relational consistency.

# 9 Role-Based Access Control

- **Organizer:** Full access (Create, Edit, Delete)
- **Viewer:** Read-only access

Permissions are enforced at both UI and logic levels.

# 10 Key Features

## 10.1 Tournament Management

Users can create, edit, and delete tournaments.

## 10.2 Team Management

Teams can be added with custom logos from the device gallery.

### 10.3 Match Management

Matches can be scheduled with:

- Teams
- Scores
- Dates

### 10.4 Leaderboard System

The leaderboard is dynamically calculated based on match results.

### 10.5 Validation

- Prevents invalid matches
- Ensures proper data input

## 11 Leaderboard System

The leaderboard follows standard football rules:

- Win = 3 points
- Draw = 1 point
- Loss = 0 points

Ranking is determined by:

1. Points
2. Goal Difference
3. Goals Scored

The leaderboard is calculated dynamically to avoid redundant data storage.

## 12 UI/UX Design

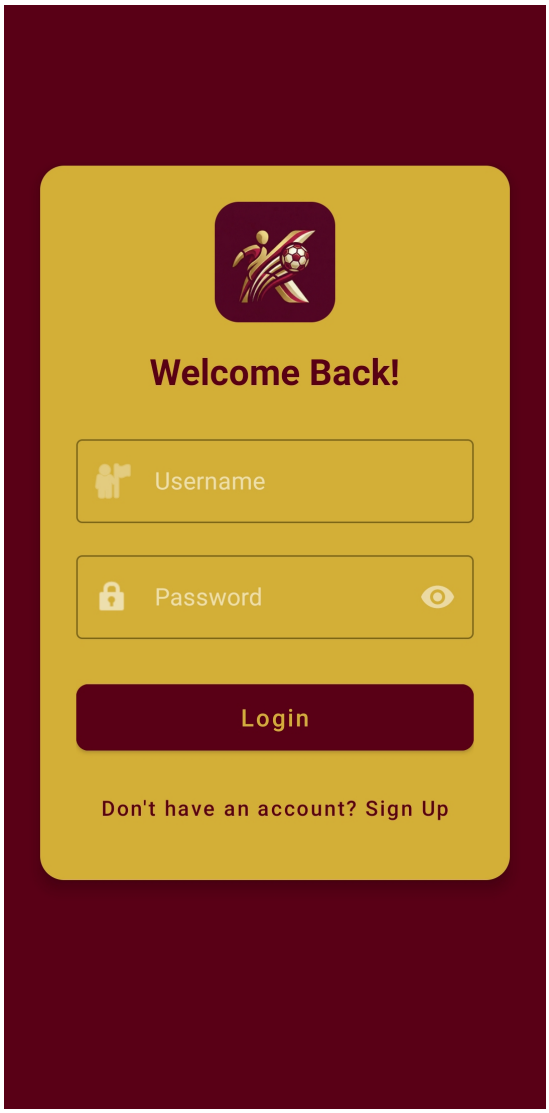
The application uses a custom Maroon and Gold theme.

- MaterialCardView for structured layouts
- RecyclerView for lists
- Consistent spacing and typography
- Smooth animations and transitions

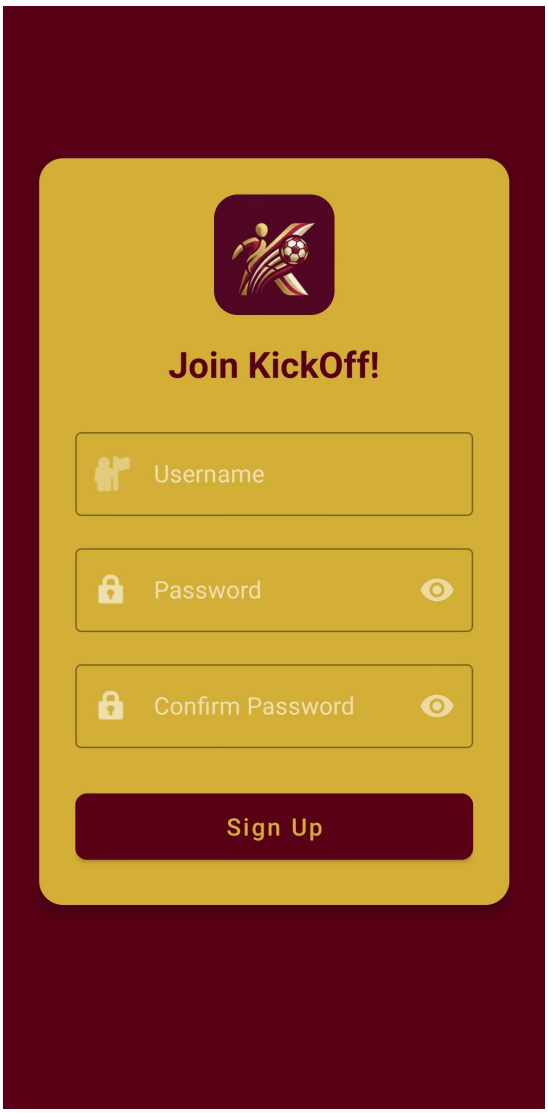
# 13 Application Screenshots

This section presents key screens of the KickOff application to demonstrate the user interface and workflow.

## 13.1 Authentication Screens



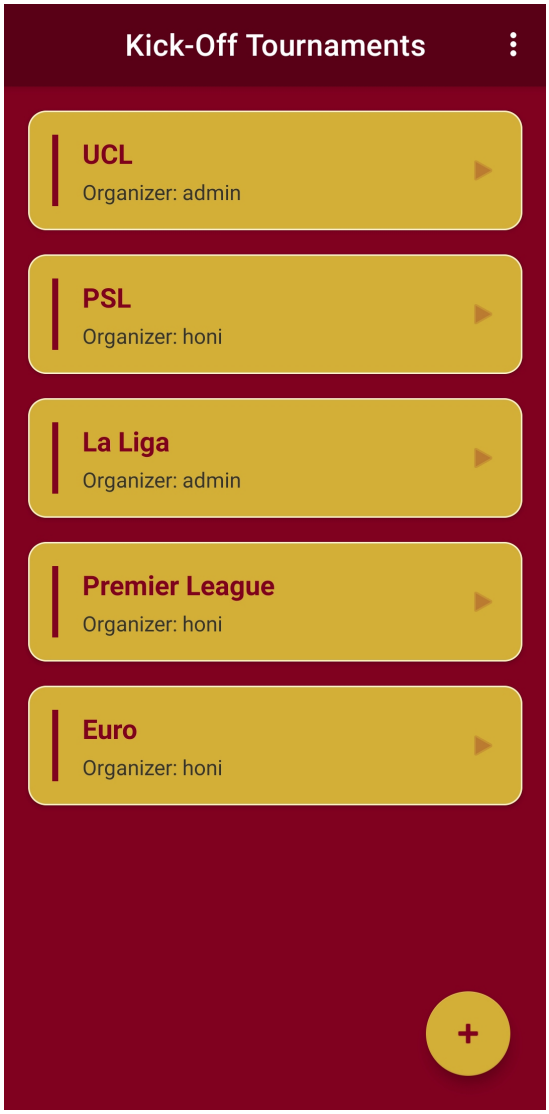

(a) Login Screen



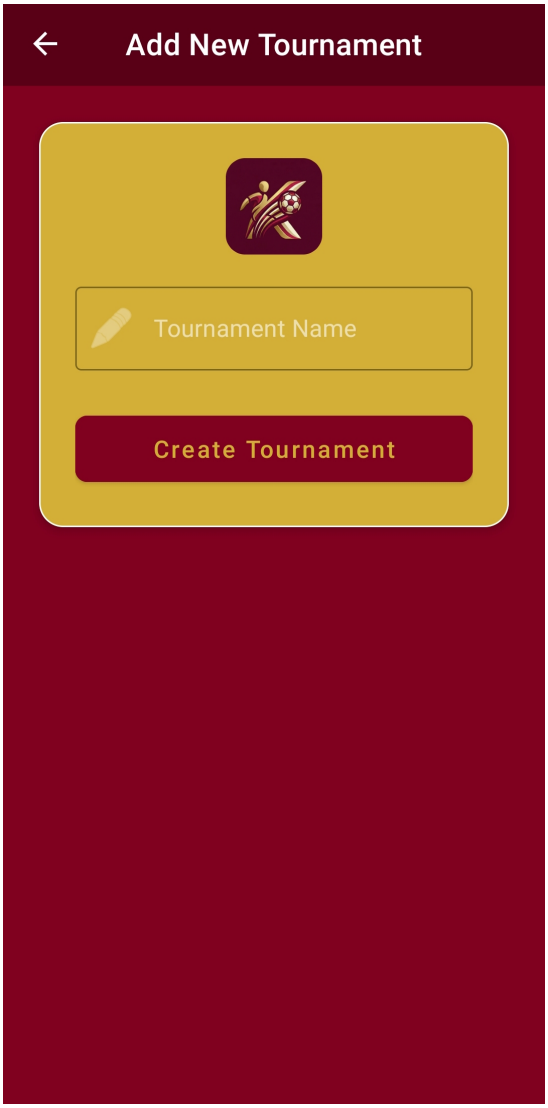

(b) Signup Screen

Figure 1: User Authentication Interfaces

### 13.2 Tournament Module



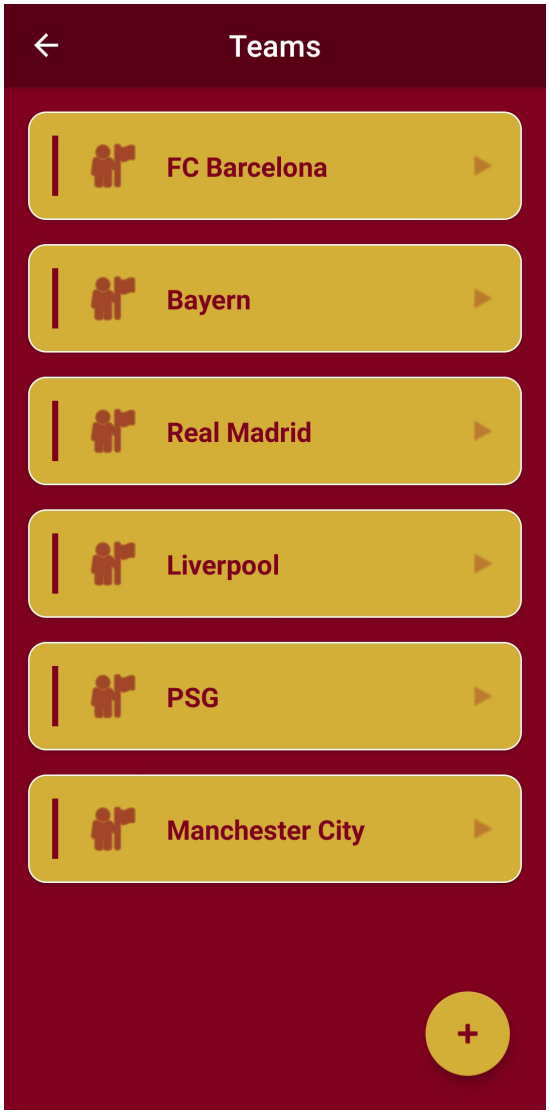

(a) Tournament List




(b) Add Tournament

Figure 2: Tournament Management

### 13.3 Team Management




(a) Team List

(b) Match List

Figure 3: Teams and Matches Screens

### 13.4 Leaderboard

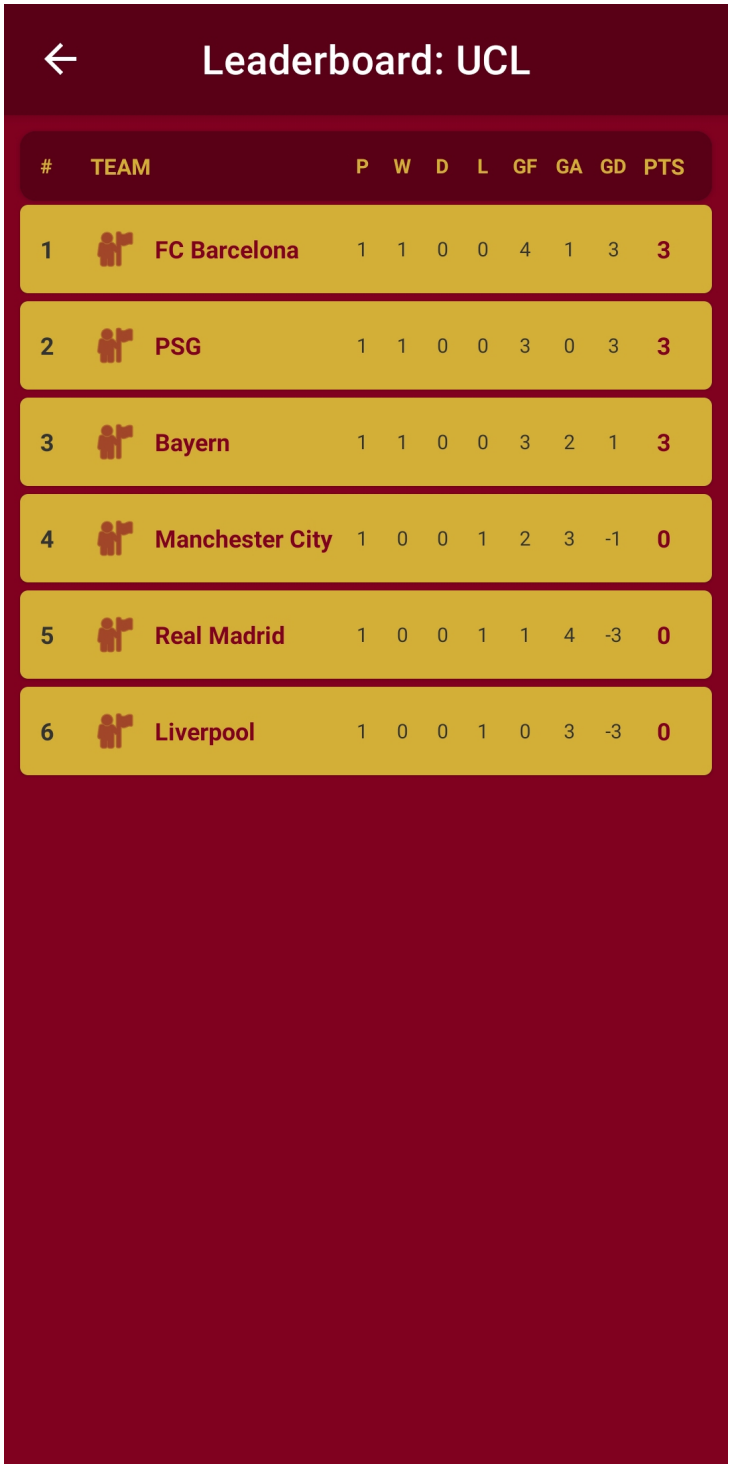



Figure 4: Dynamic Leaderboard Display

## 14 Performance Optimization

To handle large images:

- Images are downsampled before loading

- Prevents memory issues and lag

## 15 Limitations

- No cloud synchronization
- Limited scalability for very large datasets
- Uses local storage only

## 16 Future Enhancements

- Integration with Firebase or SQL database
- Real-time multi-user synchronization
- Push notifications
- Advanced analytics and statistics

## 17 Conclusion

KickOff successfully demonstrates the development of a fully functional Android application with structured data management, role-based access control, and dynamic computation.

The project highlights strong understanding of mobile application architecture, user experience design, and efficient data handling without relying on a backend system.